

CLASS 11 / NOV 3 / U07

SDF VOLUME STUDIO WORK

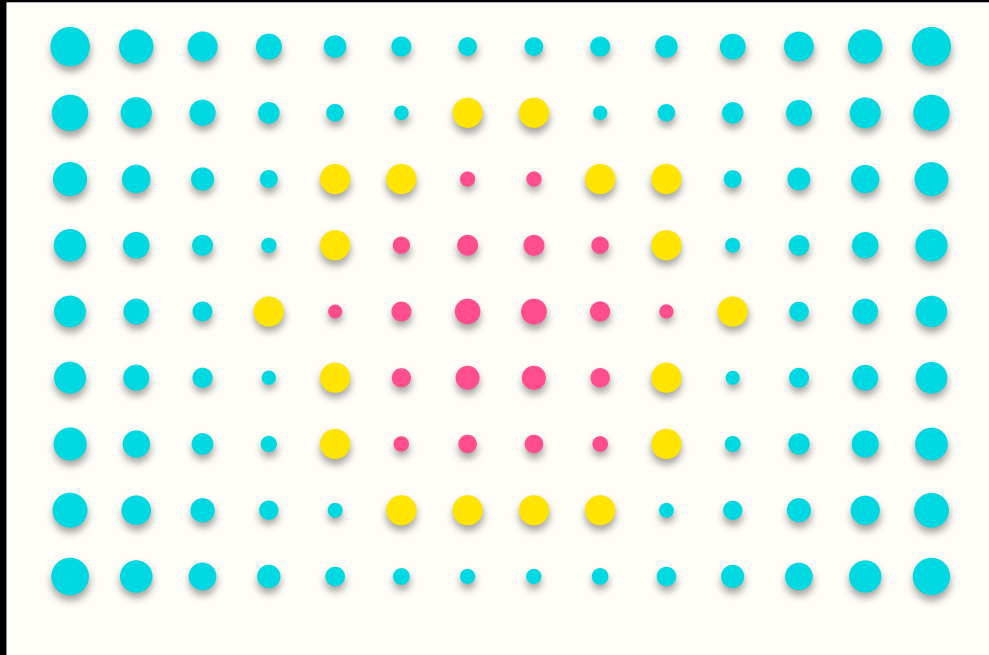
P4 CONTINUES — ONE SHAPE, WRITTEN AS A FUNCTION

P4a CHECKPOINT

VOLUMETRIC DATA AND FIELDS

A FIELD HAS A SIGN

At every point the function returns one number: how far to the surface, and on which side.



NEGATIVE

Inside the shape. The further in, the more negative.

ZERO

Exactly on the surface. This is the shape you will see.

POSITIVE

Outside the shape. The value is the distance to the surface.

$$\text{sphere}(p) = \text{length}(p - \text{center}) - \text{radius}$$

THE SURFACE IS WHERE THE FIELD EQUALS 0

PRIMITIVES ARE FUNCTIONS

Each one takes a point and returns a signed distance. Position and size are inputs.

SPHERE

$$d = \text{length}(p - c) - r$$

Distance to the center, minus the radius. The simplest exact field.

BOX

$$q = \text{abs}(p - c) - \text{half}$$
$$d = \text{distance past the nearest face}$$

Negative inside, zero on each face, positive outside.

PLANE

$$d = \text{dot}(p, n) - h$$

One side is negative, the other positive. Useful to cut or to slice.

Start with two primitives you can name and move. Expose center and size as inputs, so every variant on the sheet is one number away.

THE P2 BOOLEANS, WRITTEN ON NUMBERS

Same operands, same order rules. The operation is now a comparison of two values.

UNION

$$\min(a, b)$$

Keep the nearer surface. Inside either field counts as inside.

INTERSECTION

$$\max(a, b)$$

Keep the farther surface. Inside only where both are negative.

DIFFERENCE

$$\max(a, -b)$$

Flip b inside out, then intersect. a with b carved away.

$\max(a, -b)$ IS NOT $\max(b, -a)$. NAME THE OPERANDS ON THE SHEET, AS YOU DID IN P2a.

P4a asks for all three from the same source fields. Change nothing but the operation, so the comparison is fair.

SMOOTH BLEND, AND WHAT NOISE DOES

Two ways to move past hard Booleans. Only one of them keeps the field an exact distance.

SMOOTH MIN

```
d = smooth_min(a, b, k)
```

k = 0 gives the plain union. A larger k widens the fillet where the two shapes meet.

k is in model units. Write its value on the sheet.

NOISE

```
d = field(p) + amp × noise(p)
```

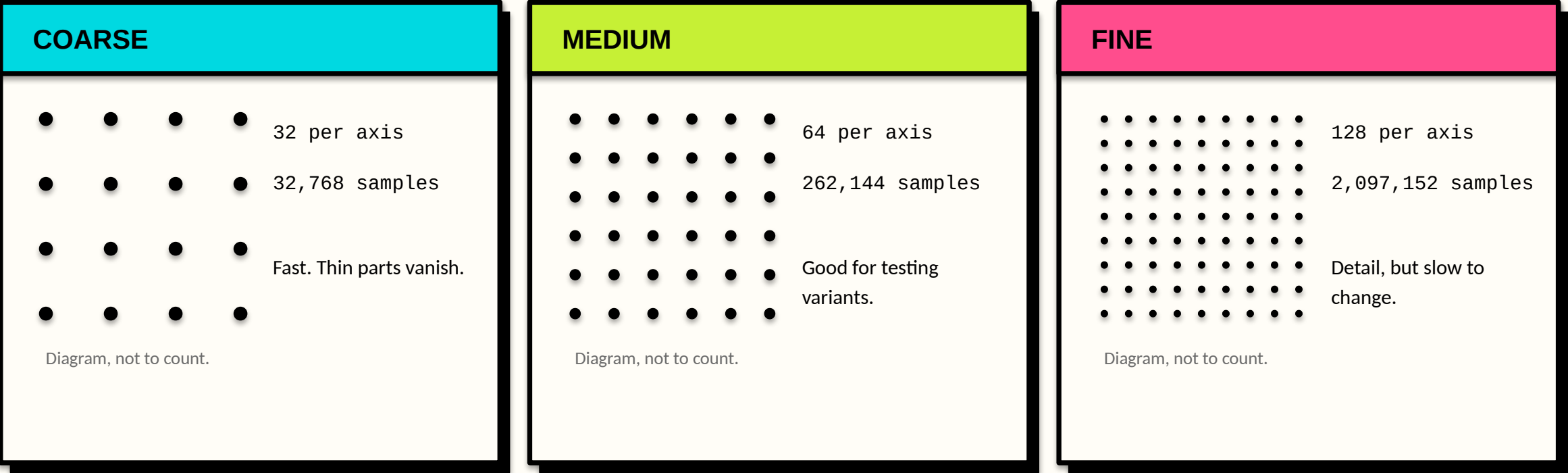
Noise changes the field, and the zero surface moves with it.

The values are no longer exact distances. Call the result an implicit field, not an SDF.

CHANGE ONE INFLUENCE AT A TIME: k, OR amp, OR NOISE SCALE — NEVER TWO IN THE SAME VARIANT.

RESOLUTION: DETAIL AGAINST COST

The field is continuous. You only ever see it at the points where you sample it.



DOUBLE THE RESOLUTION ON EACH AXIS AND THE SAMPLE COUNT GROWS EIGHT TIMES. WORK COARSE, FINISH FINE.

P4a requires the sampling resolution stated on the sheet.

SURFACE EXTRACTION

The mesh is the last step, not the design. Everything upstream is still a number per sample.

01 DEFINE

Write the field: primitives, Booleans, blend.

02 SAMPLE

Evaluate it on a grid inside stated bounds.

03 EXTRACT

Build the surface where the value crosses the threshold.

04 COMPARE

Extract at two resolutions. Keep both.

```
def field(p):  
    return smooth_min(a(p), b(p), k)  
grid      = sample(field, bounds, resolution)  
surface = extract(grid, iso=0.0)
```

STATE ON THE SHEET

Bounds · resolution · threshold (iso) · any smoothing after extraction

Keep the source fields live. The extracted mesh is what you will slice in Week 12.

P4a — TWO SHEETS, TWO JOBS

Two 17 × 11 inch sheets. One explains the field. The other argues for a volume.

SHEET 1 — THE FIELD LOGIC

- The source fields, named, with their inputs
- A diagram of the field: inside, outside, zero
- Union, intersection and difference from the same sources
- The implicit function named: density, SDF or other

SHEET 2 — THE VOLUME

- The chosen result, large
- Three controlled variants, one parameter each
- Sampling resolution and extraction settings
- The volume selected for P4b, and its spatial intention

Sheet 2 ends with a decision. If you cannot say why this volume should be sliced, you are not done with P4a.

REMINDER — P4a

Visual quality 35% · computational understanding 25% · technical execution 20% · experimentation 10% · requirements and identity 10%.

01

Use two 17 x 11 inch sheets to explain a volume built from signed distance fields (SDF) — the CSG logic from P2, applied to fields instead of meshes.

02

Compare union, intersection and difference using the same source fields. Show the chosen result and three controlled parameter variants.

03

Diagram the field logic and identify density, signed distance or another implicit function. State sampling resolution and surface extraction settings.

04

Select one volume for P4b and explain the spatial intention. Submit the editable graph and retain the source fields.

CHECKPOINT · WEEK 11 — Nov 3 — Bring an SDF diagram, a union/intersection/difference comparison and a candidate volume for slicing.

NOW — CHECKPOINT AND WORK

Rest of the class: the checkpoint first — items 1, 2 and 5 — then keep building.

✓

An SDF diagram: inside, outside and the zero surface, for your own fields

✓

Union, intersection and difference from the same two source fields

✓

One smooth-min or noise variant, with the changed value written down

✓

The surface extracted at two resolutions, both kept

✓

A candidate volume for slicing, with one sentence of intention

IF YOU CANNOT SAY WHICH SIDE IS NEGATIVE, START THERE.

BEFORE NEXT CLASS

NEXT

Week 12 (target Nov 10) — Discretizing Geometry: section planes, direction, interval and order. Watch U09 before class.

SUBMIT

P4a is due next class. Due Week 12 — Nov 10 / 14%. Two sheets as PDF, plus the editable field graph with the source fields retained.

BRING

BOOK checkpoint, Week 12 — Nov 10: one booklet spread and your contents list, for an ungraded progress check.

CHOOSE

The volume you will slice. Export or keep the extracted surface ready to cut with section planes next week.